

Qn	Solution
8(a)(i)	displacement 0 0 12 20 $t / \mu\text{s}$
8(a)(ii)	viscous force by air on oil drop, R weight of oil drop, mg
8(b)(i)	
8(b)(ii)	From (a)(ii), $R = mg$ From (b)(i), electric force $F = R + mg = 2mg$ i.e. $\frac{qV}{d} = 2mg \rightarrow q = \frac{2mgd}{V} = \frac{2(8.2 \times 10^{-16})(9.81)(0.20)}{5030}$ $= 6.40 \times 10^{-19} \text{ C}$ [Many candidates did not determine the drag force correctly, and consequently obtained an answer which was half of that required.]
8(b)(iii)	Elementary charge $e = 1.60 \times 10^{-19} \text{ C}$ Number of electrons transferred $= \frac{q}{e} = 4$ Direction of transfer is <u>to</u> the oil drop.
8(c)(i)	Force on oil drop by sphere, $F = \frac{Qq}{4\pi\epsilon_0 r^2} = ma$ $a = \frac{Qq}{4\pi\epsilon_0 r^2 m} = (9 \times 10^9) \frac{(2.0 \times 10^{-6})(1.1 \times 10^{-18})}{(0.15)^2 (8.2 \times 10^{-16})}$ $= 1.07 \times 10^3 \text{ m s}^{-2} = 1.1 \times 10^3 \text{ m s}^{-2}$

Qn	Solution
8(c)(ii)	The acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$ is less than 1% of the initial acceleration $a = 1.07 \times 10^3 \text{ m s}^{-2}$ from (c)(i). So the weight of the oil drop is negligible compared to the electric force acting on it by the sphere. The two objects are positive, so the oil drop is repelled by the sphere and it moves away from the sphere along the straight line joining the centres of the objects. It accelerates from rest with an initial acceleration of $1.07 \times 10^3 \text{ m s}^{-2}$. This acceleration decreases with time as the oil drop moves further away from the sphere, since the force that causes the acceleration decreases, being inversely proportional to the square of the separation between the oil drop and the sphere. After some time, the path starts to curve down when g is no longer negligible compared to a. [Most candidates gave a description of how the motion of the oil drop varies with time, but did not explain. The idea of the weight contributing to the path of the oil drop was not always appreciated. Some candidates brought in air resistance, and some responses stated that the oil drop ceased to move once it was far enough from the sphere. Some responses had the oil drop accelerating towards the sphere.]
9(c)(i)	The principle of superposition states that when two or more travelling waves of